EXPERIMENT 12

Implementation of Echo Client–Server Using TCP and UDP Sockets

Exp 12(a)

Date: 22/9/26

Implementing of echo client server using TCP & UDP sockets

Aim:

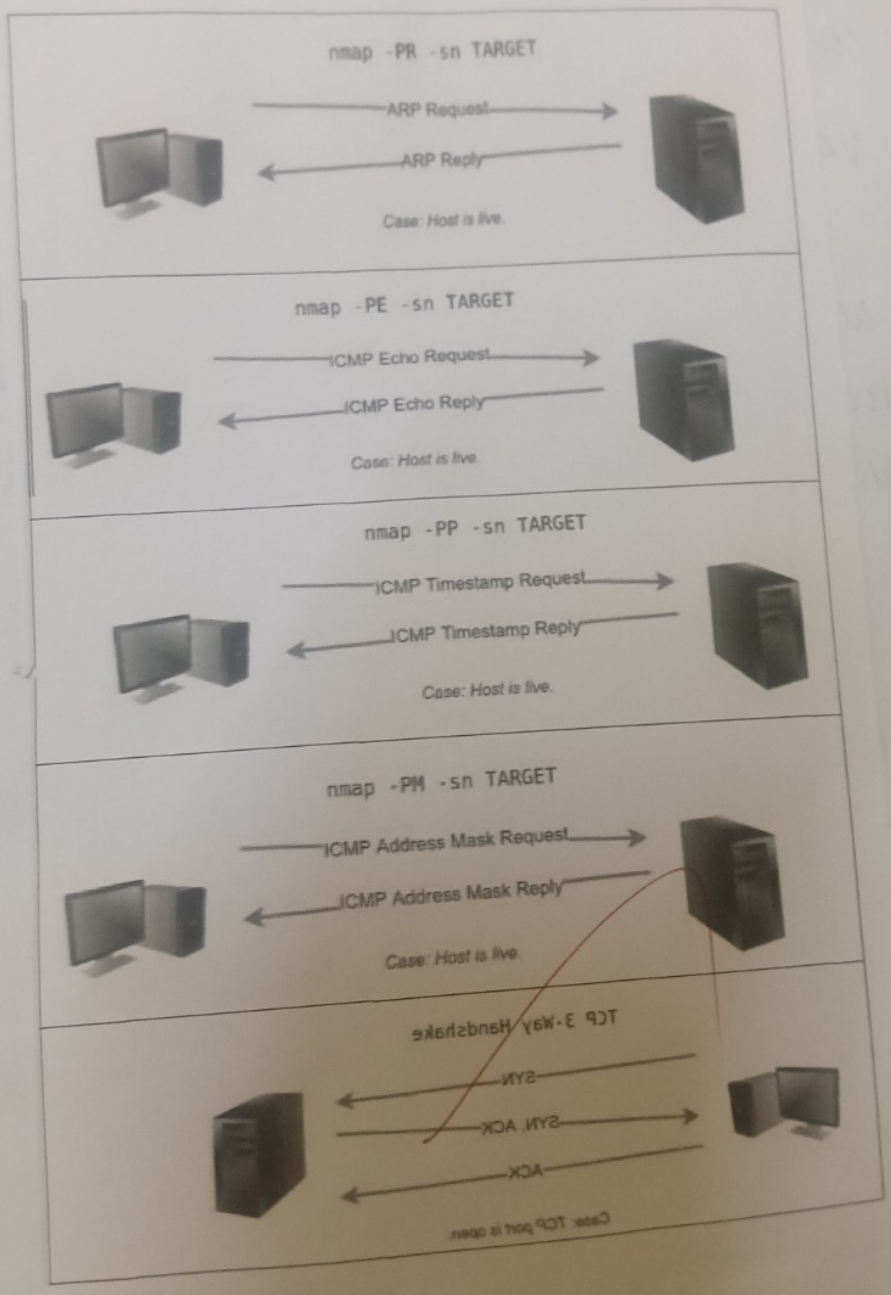
To implement the echo client server program using TCP sockets in.

Algorithm:

1) Start the TCP server
2) Accept the read message sent by client
3) send same msg back to client
4) Display echoed message and close the connection UDP echo servs
5) start the UDP servers
6) create Datagram sockets
7) send some msg back to client

UDP echo client

1) start UDP client
2) create a datagram socket
3) Read msg from user & create a datagram packet. into server
4) Display echoed msg
5) close the connection


nmap -PR -sn TARGET
ARP Request
ARP Reply
Case: Host is live.
nmap -PE -sn TARGET
ICMP Echo Request
ICMP Echo Reply
Case: Host is live.
nmap -PP -sn TARGET
ICMP Timestamp Request
ICMP Timestamp Reply
Case: Host is live.
nmap -PM -sn TARGET
ICMP Address Mask Request
ICMP Address Mask Reply
Case: Host is live.

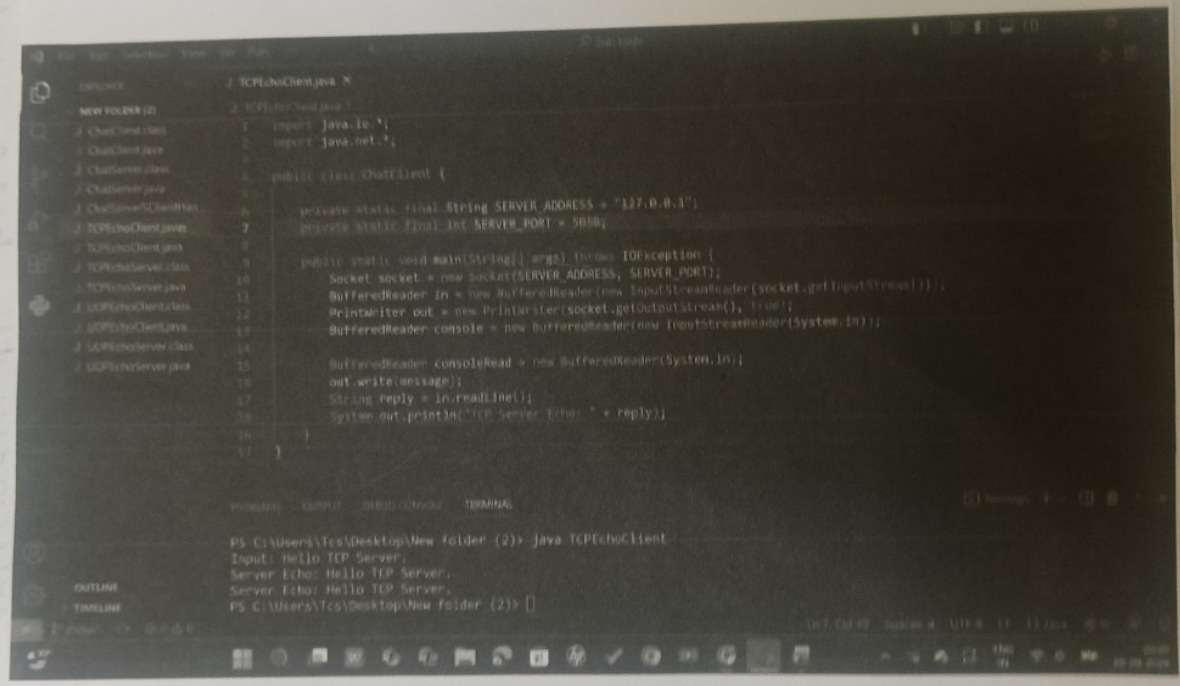
Code:

TCP Echo Server

```
import java.io.*;
import java.net.*;
public class TCP Echo Server {
  public static void main() {
    ServerSocket server = new ServerSocket
    Socket = server.accept()
    BufferedReader in = new BufferedReader(
    Socket.close())
    server.close())
  }
}
```

TCP Echo client:

```
import java.io.*;
import java.net.*;
public class TCP.Echo client {
  public static void main() {
    Socket Socket = new Socket ("local
                                HOST", 5000)
    print (Writer out = new print (true)
    while(true){
      System.out.print ("Enter a message")
      socket.close ()
  } }
}
```


TCPEchoClient.java
NEW FOLDER (2)
ChatClient.class
ChatClient.java
ChatServer.class
ChatServer.java
ChatServerClientHan...
TCPEchoClient.java
TCPEchoClient.java
TCPEchoServer.class
TCPEchoServer.java
UDPEchoClient.class
UDPEchoClient.java
UDPEchoServer.class
UDPEchoServer.java
import java.io.*;
import java.net.*;
public class ChatClient {
private static final String SERVER_ADDRESS = "127.0.0.1";
private static final int SERVER_PORT = 5050;
public static void main(String[] args) throws IOException {
Socket socket = new Socket(SERVER_ADDRESS, SERVER_PORT);
BufferedReader in = new BufferedReader(new InputStreamReader(socket.getInputStream()));
PrintWriter out = new PrintWriter(socket.getOutputStream(), true);
BufferedReader console = new BufferedReader(new InputStreamReader(System.in));
BufferedReader consoleRead = new BufferedReader(System.in);
out.write(message);
String reply = in.readLine();
System.out.println("TCP Server Echo: " + reply);
}
}
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PS C:\Users\Tcs\Desktop\New folder (2)> java TCPEchoClient
Input: Hello TCP Server.
Server Echo: Hello TCP Server.
Server Echo: Hello TCP Server.
PS C:\Users\Tcs\Desktop\New folder (2)>
OUTLINE
TIMELINE

Result:

Thus, the echo client-server app was successfully implemented TCP & UDP sockets the message sent to the client was successfully echoed by the server